

CUSTOMER FEATURE GUIDE

Veyron

Kubernetes-native VM command center.

Declarative VM provisioning, day-2 operations, and a Mission Control dashboard for KubeVirt — forged in Rust.

by ZyvorAI Labs

What is Veyron?

Veyron turns verbose KubeVirt YAML into a single, opinionated command center for virtual machines on Kubernetes. It ships four surfaces over one Rust core — a CLI, a k9s-style TUI, a CloudOS web dashboard, and a REST/WebSocket API — plus a Go operator that makes VMs GitOps-native through the VeyronVM custom resource. From a 44-image template catalog and multi-VM blueprints to browser consoles, snapshots, live migration, SOC detections, and an AI copilot, Veyron covers the full VM lifecycle from create to retire.

44

OS templates across 15 families

40+

dashboard pages

4

surfaces: CLI, TUI, Web, API

8

resource profiles

What's inside this guide

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VM Provisioning & Templates

From a blank cluster to a running fleet without hand-writing manifests.



Declarative VM Builder

Create KubeVirt VirtualMachines from the CLI, wizard, or dashboard with CPU, memory, disk, storage class, and cloud-init.

Skip error-prone YAML — describe the VM you want and Veyron generates a validated manifest.



44-Image OS Catalog

Launch from 44 curated templates spanning 15 OS families — Ubuntu, Fedora, Debian, RHEL, Rocky, AlmaLinux, Windows, and BSD.

Battle-tested base images with sensible defaults for every common guest.



Multi-VM Blueprints

Deploy whole stacks — LAMP, a Kubernetes cluster, 3-tier apps, CI/CD — as one dependency-ordered blueprint.

Stand up an entire environment with a single deploy instead of provisioning VMs one at a time.



Resource Profiles

Apply prebuilt sizing profiles (dev, test, prod, high-perf, microservice) or author your own with custom CPU topology.

Consistent, right-sized VMs across teams without re-deciding cores and memory each time.



Interactive Wizard & Batch

Walk through guided VM creation, or provision many VMs at once from a batch YAML/JSON file.

One prompt-driven flow for newcomers, one file for bulk fleet builds.



Clone, Export & Dry-Run

Duplicate an existing VM, export any VM as a manifest, or preview the generated resource before it touches the cluster.

Reproduce known-good VMs and review exactly what will be applied first.

Templates resolve from an embedded catalog for offline CLI use and from cluster VMTemplate/VMProfile CRDs for GitOps — the operator and CLI agree on names.

Consoles & Guest Access

Reach any VM directly from the browser or terminal — no virtctl timeouts.



Browser VNC Console

Open a graphical VNC session to any VM over a direct Kubernetes WebSocket, right in the dashboard.

[Full desktop access without VPNs, jump hosts, or virtctl port-forwards that time out.](#)



Serial Console & Logs

Attach to a VM's serial console and stream logs from its virt-launcher pod with follow and tail.

[Diagnose boot problems and kernel panics where a graphical console can't reach.](#)



RDP Screen Sharing

Expose and connect to Windows guests over RDP through the dashboard's Screen Sharing surface.

[Native Windows remote-desktop access managed from the same console as your Linux VMs.](#)



SSH & Key Injection

SSH into running VMs and inject public keys via KubeVirt accessCredentials, propagated through the guest agent or configDrive.

[Provision secure key-based access to a VM without rebuilding or manual guest edits.](#)



Guest Agent Signals

Read QEMU guest-agent data — in-guest filesystem usage, network info, and health signals — from CLI or dashboard.

[See what's actually happening inside the guest, not just what Kubernetes reports outside it.](#)

Snapshots & Data Protection

Application-consistent point-in-time copies, backups, and disaster-recovery plans.



VM Snapshots

Create, list, inspect, delete, and restore KubeVirt VirtualMachineSnapshots, including in-place restore.

Roll a VM back to a known-good state in seconds before risky changes.



App-Consistent Freeze

Quiesce guest filesystems with fsfreeze via the QEMU guest agent before snapshotting, then thaw them.

Snapshots that databases and apps can actually be restored from — not just crash-consistent copies.



VM Backups

Take full, incremental, or differential backups with gzip/zstd/lz4 compression and integrity verification.

Portable, verifiable VM copies with scheduled full/incremental rotation.



Backup Schedules

Define hourly, daily, weekly, or monthly backup schedules scoped to specific VMs or the whole fleet.

Set-and-forget protection so no VM drifts out of your backup window.



Velero & DR Plans

Surface Velero Backup/Restore status and author disaster-recovery plans with dry-run execution and failover.

Cluster-level recovery readiness alongside per-VM backups, tested before you need it.

Migration & High Availability

Move workloads between nodes and keep critical VMs running through maintenance.



Live & Offline Migration

Migrate VMs across nodes using live, offline, or post-copy strategies, with a plan preview before executing.

Rebalance load or evacuate hardware without downtime for migratable workloads.



Node Evacuation & Drain

Cordon, uncordon, and evacuate nodes with controlled parallel migrations, timeouts, and status tracking.

Take a node down for maintenance while VMs safely relocate themselves.



VM High Availability

Configure per-VM HA priority and eviction strategy (live-migrate, shutdown, none) and view HA status.

Guarantee that your critical VMs are the ones that survive node failures.



Migration Tracking & Policies

Watch live migration progress and manage KubeVirt migration policies from the scheduling surface.

Know exactly where a migration is and enforce fleet-wide migration guardrails.

Storage & Disks

Grow, hotplug, and monitor VM disks without downtime.



Online Disk Expansion

Expand VM disks and resize PVCs, with a plan preview and generated in-guest filesystem-grow scripts.

[Add capacity to a running VM instead of rebuilding it when it fills up.](#)



Volume Hotplug

Attach and detach PVCs on a running VM via virtctl addvolume/removevolume.

[Give a live VM more storage on demand — no restart required.](#)



Disk Health & Usage

Report per-VM disk health and usage statistics, sorted by utilization, size, or free space.

[Spot the disks about to fill before they take an application down.](#)



Storage Pools & PVCs

Browse PVCs, storage classes, and storage pools across the cluster from the Storage page.

[One inventory of every volume backing your VM fleet.](#)

Networking

Per-VM connectivity, policy enforcement, and network intelligence.



Interfaces & Bandwidth

List VM network interfaces, monitor per-interface bandwidth in watch mode, and analyze traffic and top talkers.

[See which VM is saturating the network and on which interface.](#)



Multus Multi-Network

Attach additional Multus networks to a VM and manage NetworkAttachmentDefinitions.

[Give VMs the multiple NICs and VLANs that real workloads need.](#)



Network Policies & Egress

List and inspect NetworkPolicies, and apply a default internet-egress policy on VM create (opt-out available).

[VMs come locked down by default instead of wide open to the internet.](#)



Cilium & PacketWolf

View Cilium agent status and policies, with an optional PacketWolf network-intelligence integration.

[Deeper network visibility for clusters running Cilium or the PacketWolf brain.](#)

AI Copilot — Ask Zeus

Plain-language diagnosis with cluster evidence and one-click fixes.



Ask Zeus Assistant

Ask VM questions in natural language (🗨️) and get root cause, cluster evidence, and recommended actions.

[Skip the stack traces — describe the problem and get a grounded answer.](#)



Veyron Doctor

Generate a per-VM health score with prioritized issues drawn from live KubeVirt/Kubernetes state.

[A quick triage read on any VM before you dig into it manually.](#)



Scheduling Explainer

Translate a Pending or Unschedulable VM into a human-readable reason and fix.

[Understand why a VM won't start without decoding scheduler events yourself.](#)



Error Explainer & YAML Builder

Turn cryptic K8s/KubeVirt errors into plain English and generate cluster-validated VirtualMachine YAML.

[Get unblocked on obscure errors and hand off valid manifests in one step.](#)



Lens Advisors

Purpose-built advisors — Network Lens, Storage Doctor, Backup Advisor, Cost Advisor, Security Sentinel — inspect a VM or the whole fleet.

[Domain-specific expertise on connectivity, bloat, coverage gaps, spend, and exposure.](#)

The v1 assistant is deterministic and works without any external LLM by composing real cluster data; connect an OpenAI-compatible model (VEYRON_AI_*) to enable richer conversational answers.

Security & Compliance

Detections, hardening, policy enforcement, and an audit trail for the VM fleet.



SOC & Detections

Run a security operations center over VM activity — detections, threat hunts, attack-surface mapping, and SIEM export.

[Turn VM events into actionable security signal, pushed to Elastic or your SIEM.](#)



Hardening Profiles

Scan VMs and apply hardening baselines — CIS, STIG, PCI-DSS, NIST — with verify-only mode.

[Bring VMs up to a recognized security standard on demand.](#)



Compliance Frameworks

Check VMs against PCI-DSS, HIPAA, SOC 2, ISO 27001, GDPR, NIST, and CIS, and produce reports.

[Evidence for auditors without manual per-VM checklists.](#)



Policy Enforcement

Define VeyronPolicy CRDs with CEL deny rules so violating VMs never make it into the cluster.

[Stop misconfigured or non-compliant VMs at admission instead of cleaning up after.](#)



RBAC, Audit & SSO

Inspect cluster RBAC roles and bindings, review a namespace-aware audit trail, and sign in via OIDC SSO.

[Know who did what, and gate access to identity your org already trusts.](#)

Cost & FinOps

Understand, budget, and trim what your VM fleet spends.



Cost Analysis

Break down fleet and per-VM cost by namespace, team, or project over configurable time windows.

[See where the money goes across every VM you run.](#)



Budgets & Alerts

Create budgets scoped to global, namespace, team, or project with configurable alert thresholds.

[Get warned before spend blows past what you planned for.](#)



Waste & Optimization

Surface idle, oversized, storage, and snapshot waste, and get prioritized cost-optimization recommendations.

[Reclaim spend from the VMs and volumes nobody is using.](#)



Cost Forecasting

Project 7/30/90-day costs and compare them against a budget.

[Plan next quarter's VM spend before it arrives.](#)

Observability & Monitoring

Live metrics, hotspots, SLOs, and stack discovery across the fleet.



Live VM Monitoring

Stream live CPU/memory/disk stats for a VM, compare multiple VMs, and rank the top resource consumers.

[Watch a VM in real time and instantly find the fleet's heaviest workloads.](#)



Hotspots & Heatmap

Aggregate workload requests and render a resource heatmap to expose CPU and memory pressure.

[Spot the nodes and VMs running hot at a glance.](#)



SLOs & Alerts

Track fleet availability SLO objectives and read namespace-aware warning-event narratives and alerts.

[Turn raw Kubernetes events into an availability story you can act on.](#)



Stack Discovery

Auto-detect the observability stack — Prometheus, Grafana, Loki, Jaeger — and deep-link into it.

[Jump from Veyron straight into the monitoring tools you already run.](#)

GitOps & Operator

CR-native VMs, drift detection, and infrastructure-as-code delivery.



VeyronVM Operator

A Go controller-runtime operator reconciles VeyronVM and VeyronBlueprint custom resources into live KubeVirt VMs.

Manage VMs the Kubernetes-native way — declare them and let the operator converge.



GitOps Export

Export every VM in a namespace as VeyronVM CRD manifests with a single command.

Capture a running fleet as version-controlled YAML for Argo CD or Flux.



Drift Detection

Compare live VMs against their desired CRD state and surface drift as badges, filters, and operator insights.

Know the moment a VM diverges from what Git says it should be.



Helm & Terraform

Install via Helm charts and provision through a dedicated Terraform provider.

Fit Veyron into whatever IaC pipeline your platform team already uses.



Catalog CRDs

Publish embedded templates and profiles to the cluster as VMTemplate/VMProfile CRDs and keep them in sync.

One authoritative, GitOps-managed catalog shared by CLI, operator, and dashboard.

Platform & Surfaces

One Rust core, four ways to drive it, across many clusters and tenants.

Surface	Best for
Web dashboard (CloudOS)	Visual operations, consoles, guided fixes
CLI (veyron)	Scripting, automation, CI/CD pipelines
TUI	Fast keyboard-driven cluster navigation
REST / WebSocket API	Custom tooling and integrations



CloudOS Dashboard

A macOS-style Mission Control web shell with 40+ pages, Spotlight (⌘K), a dock, Finder sidebar, and Fix-it flows.

A polished, discoverable home for operators who'd rather click than memorize flags.



TUI & CLI

A k9s-style terminal UI with themes and dialogs, plus a full-featured headless CLI for scripting.

Fast keyboard-driven control at the terminal, and automatable commands for pipelines.



REST & WebSocket API

A Rust Axum API server (49 routes) with embedded OpenAPI and WebSocket console tunnels.

Build your own tooling on the same API the dashboard and CLI use.



Multi-Cluster & Tenancy

Switch between kubeconfig contexts from a cluster bar and carve the fleet into tenant workspaces with quotas and billing tags.

Operate many clusters and isolate teams from a single control plane.



Automation & Integrations

Define automation rules (manual, schedule, event, metric triggers) and probe optional integrations from one page.

Codify routine day-2 responses and see which backends are wired at a glance.

GETTING STARTED

Your first 30 minutes

A suggested path to value on day one.

- 1 Build and install**
Clone the repo and run `cargo build --release` to produce the veyron binary.
- 2 Create your first VM**
Run `veyron create --template ubuntu-22.04 --name web-01` to launch a VM from a template.
- 3 Deploy a stack**
Run `veyron blueprint deploy lamp --namespace dev` to stand up a full multi-VM blueprint.
- 4 Open Mission Control**
Run `veyron serve` and browse to `https://localhost:8080` for the CloudOS dashboard and API.
- 5 Go GitOps**
Install the operator from `charts/veyron-operator` and run `veyron gitops-export` to capture VMs as VeyronVM manifests.

Support & contact

Veyron is developed by **ZyvorAI Labs**. For onboarding, a proof-of-concept, or a guided demo, contact your account team at info@zyvor.dev.

Good to know: Veyron is a control plane over KubeVirt and Kubernetes and depends on those APIs being present. Several dashboard pages are honest about their limits: costs are estimates unless OpenCost is wired, security posture is configuration-derived rather than scanner-backed, and forecasting, observability counts, and scheduling metrics are heuristic. Optional integrations (OpenCost, Trivy, Prometheus, PacketWolf, Elastic/SIEM, an OpenAI-compatible LLM) unlock deeper data when configured.

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